

After Acceptance: A Claim Graph for Residual Languages and Weird-Machine Claims, Calibrated on eBPF

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Abstract—Language-theoretic security models an input boundary as a recognizer for a formal language and downstream processing as interpretation. At a program-verifier boundary, the verifier is an outer recognizer of program artifacts; an accepted artifact may in turn interpret data and drive runtime-operation words, while any computed report is a separate, explicitly declared abstraction. We introduce a five-node claim graph separating artifact acceptance (A), a same-suffix causal state distinction (C), bounded programmability (P), output-witnessed computed-report non-factorization (R), and a policy/threat obligation (W). A future-observation quotient and behavioral factorization criterion distinguish the accepted-artifact-indexed causal language from the report-relative residual language. The criterion is report-instance-relative, applies only under an explicit unique-cell report map on the chosen context fiber, and does not assert that every safety-sound incomplete verifier hosts a weird machine.

We give proof obligations for a uniformly controlled, observable, resettable gate and one fixed accepted interpreter over a bounded circuit-description domain. Prefix induction yields bounded composition under exact scheduling, admissibility, serialization, and frame-preservation premises. In a Linux eBPF calibration, a dedicated preallocated non-LRU two-entry hash map realizes NAND: reset leaves one sentinel entry; input bits select existing- or fresh-key updates; and the second update’s success predicate supplies the output. One fixed accepted program consumes canonical NAND DAGs with at most 64 inputs, 512 gates, and 578 live wires.

A source-snapshotted Linux/aarch64 run covers named and random circuits, joint bounds, serial reuse, mechanism controls, and malformed descriptors. A separate author-run semantic auditor reconstructs descriptors and circuit semantics, and a self-issued manifest checks bundle integrity. This interpreter carrier records A, gives a conditional C witness under the declared concrete-service/no-interference contract, and supports P under additional implementation premises; it establishes neither R nor W. A fixed auxiliary executable instance establishes $R(M_{linux_r_aux_v1})$ for its own computed report. Separately, a local stock-Linux/aarch64 experiment on a different accepted XDP object captures a successful exact-level-0 `states_equal` check followed by an `is_state_visited` prune, binds the two recorded prefixes to one retained-state representative, and records different observations under one common suffix. Embedded in a retrospectively declared, hash-bound two-history carrier with an author-declared operational prune-report, those records satisfy Definition 2 for that tuple. We describe this result as a trace-local operational-prune-relative certificate, not as evidence that Linux specifies a complete functional report or as a theorem over general Linux executions. The carriers are not combined: neither R result is linked to the interpreter’s P evidence, and no result establishes W or a weird machine.

Index Terms—language-theoretic security, recognizers, resid-

ual languages, weird machines, program verification, abstract interpretation, eBPF

I. INTRODUCTION

Language-theoretic security treats an input boundary as a language-recognition boundary: downstream processing is interpretation, and the input supplies the interpreted program [1]–[4]. We use the same recognizer-shaped model for program verification. Our question begins after acceptance. A verifier accepts a program artifact, but the accepted program can still drive a language of stateful helper and service operations. Which claims are needed before that downstream language can be called residual, programmable, shape-induced, or a weird machine? In this layered view, acceptance, interpretation, and report abstraction answer different questions: membership in L_V , behavior in I , and grouping by Report_V , respectively.

The artifact language and the post-acceptance operation language have different carriers. A verifier can recognize bounded execution, typed helper use, and memory-access discipline without certifying the complete relation implemented through every documented runtime service. The accepted program may select operations, retain service state, observe returns, reset state, and compose effects. None of this alone implies a verifier defect. It first establishes a downstream operation language indexed by an accepted artifact.

The paper organizes the evidentiary burden as a claim graph:

Each row in Table I has its own artifact, execution carrier, report interface, and evidence type. The `wm_circuit` row carries the paper’s A/C/P construction. The auxiliary row supplies a positive R result only for a custom report-producing recognizer. The M_K row supplies a retrospective, trace-local R certificate only for the declared `rac_single` operational prune-report tuple. No column may be read by combining cells across rows, and no evaluated row establishes W.

Nodes C and R must not share one name. We call the accepted-artifact-indexed, same-suffix language L_{causal} . A word enters the report-relative residual language L_{res}^R only with an actual computed-cell collision under the admissibility conditions below. P and R branch after C. A policy-level weird machine requires P, W, linkage to the same encoded computation, and unintendedness; a *contract-shape-induced recognizer-relative* classification additionally requires R, documented semantics, report conformance, and granularity

TABLE I
CARRIER-BY-NODE EVIDENCE STATUS.

Carrier	A	C	P	R	W	Evidentiary interpretation
<code>wm_circuit</code>	recorded	conditional under the declared service contract	supported under source/object, serialization, and frame premises	not established	not established	construction plus audited regression evidence; no computed-cell extractor at the interpreter frontier
$M_{linux_r_aux_v1}$	finite-model acceptance	established within that tuple	not established	established for its custom report	not established	executable finite-model certificate; no refinement or bisimulation claim to stock Linux
M_K (<code>rac_single</code>)	recorded for the frozen tuple	witnessed only in the declared two-history relation	not established	retrospective certificate for an author-declared prune-report	not established	real prune capture plus normalized checks and a manuscript-level wrapper; trace-local, not a Linux functional-report contract

evidence. The graph turns “weird machine” from a rhetorical label into explicit, auditable obligations that support both positive certificates and principled non-classification.

Programmability is a separate constructive problem. A state distinction may be dead, uncontrolled, one-shot, or impossible to compose. We therefore require a program-visible readout, one uniform input dispatcher, a reset to a canonical class, and one fixed accepted interpreter with exact scheduling and frame preservation over an independently declared bounded descriptor domain. The resulting composition theorem is a proof-obligation decomposition, not a universal necessity theorem.

Our calibration case is Linux eBPF. A fixed artifact uses a dedicated two-entry hash map as a saturating resource. After reset, one sentinel is live. A zero bit updates the sentinel; a one bit inserts a fresh input-specific key. The second input-conditioned update succeeds except on input (1,1), yielding NAND. Ordinary bytecode still validates descriptors, selects keys, routes wires, controls the loop, and stores outputs. A host parser normalizes textual WMC1 into maps, while the verifier accepts only the fixed eBPF artifact.

The interpreter case is deliberately diagnostic. It records A, gives a conditional C witness under the declared service contract, and supports P under stated implementation premises, even though ordinary bytecode easily expresses the same function. It also shows why P cannot be promoted to R or combined with an absent W obligation. Its retained verifier log is not a computed-cell extractor, and the offline run supplies no violated policy. The separate M_K experiment supplies an actual prune-cell extractor for `rac_single`, not for the interpreter.

The paper contributes:

- 1) **A recognizer–interpreter–report claim graph and typed languages.** We distinguish accepted artifacts, causal runtime words, report-relative residual words, bounded programmability, and policy/threat obligations. A future-observation quotient and factorization criterion make node R testable, while countermodels show that the branches do not collapse.
- 2) **A bounded composition argument.** E1–E3 and the precise E4-D interpreter obligations isolate local gate

correctness, reset, scheduling, and global frame conditions. Prefix induction proves the functional result; safety is a separate optional premise.

- 3) **An eBPF calibration case.** A saturating-rank NAND basis and fixed interpreter target $D_{64,512}$. The recorded suite covers named and fixed-seed random DAGs, deep and joint boundaries, a zero-gate case, serial reuse, mechanism controls, and malformed descriptors.
- 4) **Carrier-separated R evidence.** A fixed auxiliary recognizer/service tuple emits a report and checks a finite report-cell collision with four negative controls. A separate stock-Linux capture records a real exact-level-0 prune and a common suffix with different outcomes. The latter becomes an R instance only after retrospective embedding in the explicitly finite `rac_single` carrier and author-declared operational prune-report of Section 5.7. We therefore claim a trace-local operational certificate for that frozen tuple, not a general Linux report result; neither R instance is linked to the interpreter’s P evidence.

The empirical scope is one privileged, offline Linux/aarch64 kernel build, two claim-bearing accepted eBPF artifacts, three additional accepted interpreter-control variants, and one accepted auxiliary custom-recognizer artifact. The result is not a general Linux report-opacity theorem, concurrent deployment result, artifact-parametric compiler, unbounded machine, vulnerability, or policy-level weird machine.

II. RECOGNIZER-RELATIVE RESIDUAL LANGUAGES

A. Recognition, execution, and report

Let V be a recognizer over program artifacts and I the concrete execution semantics over state carrier Σ_I and trace carrier $\mathcal{T}_I$, including the instruction engine, helpers, stateful services, and relevant environment. Thus $\text{Tr}_I(P) \subseteq \mathcal{T}_I$. The accepted artifact language is

$$L_V = \{P \mid V(P) = \text{accept}\}. \quad (1)$$

For an optional set $\text{Safe} \subseteq \mathcal{T}_I$ of permitted concrete traces, the boundary is safety-sound when

$$\forall P \in L_V. \text{Tr}_I(P) \subseteq \text{Safe}. \quad (2)$$

Safety soundness is a premise, not an inference from one successful load. It is also distinct from completeness of an abstract transformer. Abstract interpretation relates concrete sets and abstract elements through abstraction and concretization maps [5], but a verifier's accepted language, its computed report, transfer soundness, and completeness are different judgments. In particular, the best abstraction of a singleton need not be the analyzer-computed cell at a joined control frontier.

When a report is in scope, let $\text{Report}_V(P, \ell) \subseteq A_\ell$ be the finite set of abstract cells actually computed at frontier ℓ , with a declared concretization $\gamma_\ell : A_\ell \rightarrow \mathcal{P}(\Sigma_I)$. Frontier coverage requires

$$\text{Reach}_I(P, \ell) \subseteq \bigcup_{a^\# \in \text{Report}_V(P, \ell)} \gamma_\ell(a^\#). \quad (3)$$

Two concrete states are jointly covered only when one computed cell contains both. This point rules out a common but invalid shortcut: showing that two values have similar printed log text does not identify a computed abstract cell, a concretization, or joint coverage.

B. Causal words

For $P \in L_V$ and frontier ℓ , let $\Sigma_{\text{op}}(P)$ be a finite alphabet of program-controllable runtime operations, and let $W_{\text{run}}^\ell(P) \subseteq \Sigma_{\text{op}}(P)^*$ be the words accepted code can execute from ℓ . Calling every such word residual would rename ordinary trace semantics, so node C uses a same-suffix causal test without yet asserting report omission.

Fix before comparison an observation contract

$$K_{\text{obs}} = (\rho_{\text{obs}}, \text{Obs}, \text{Slice}, \text{Env}). \quad (4)$$

Here $\rho_{\text{obs}} : \Sigma_I \rightarrow R_{\text{obs}}$ selects the candidate state, $\text{Obs} : \mathcal{T}_I \rightarrow O_{\text{obs}}$ projects a complete concrete trace, Slice assigns to each word w a typed context projection $\text{ctx}_w : \Sigma_I \rightarrow C_w$, and Env is the declared set of fixed environment instances. The context conservatively contains every non-selected component read by the suffix or observer; an environment instance fixes the relevant resource configuration, schedule, nondeterminism, and external-interference choice.

Write $\llbracket w \rrbracket_{I,e}(\sigma) = (\tau, \sigma')$ for a defined, terminating suffix execution in environment e . The contract is *sound for w on $X \subseteq \Sigma_I$* when, for every $e \in \text{Env}$ and $\sigma, \sigma' \in X$ satisfying

$$\rho_{\text{obs}}(\sigma) = \rho_{\text{obs}}(\sigma') \quad \text{and} \quad \text{ctx}_w(\sigma) = \text{ctx}_w(\sigma'), \quad (5)$$

the two suffix executions have the same definedness, and, whenever $\llbracket w \rrbracket_{I,e}(\sigma) = (\tau, \sigma_1)$ and $\llbracket w \rrbracket_{I,e}(\sigma') = (\tau', \sigma'_1)$, then $\text{Obs}(\tau) = \text{Obs}(\tau')$. Thus $(\rho_{\text{obs}}, \text{ctx}_w, e)$ contains every factor that may affect the declared observation. Without this noninterference condition no C witness is admitted.

Definition 1 (causal state-mediated word family). A word w is causal at (P, ℓ) when $P \in L_V$, $w \in W_{\text{run}}^\ell(P)$, K_{obs} is sound for w on $\text{Reach}_I(P, \ell)$, and there are $e \in \text{Env}$ and

$\sigma_0, \sigma_1 \in \text{Reach}_I(P, \ell)$ for which $\llbracket w \rrbracket_{I,e}(\sigma_i) = (\tau_i, \sigma'_i)$ are both defined and terminating, and

$$\begin{aligned} \text{ctx}_w(\sigma_0) &= \text{ctx}_w(\sigma_1), \\ \rho_{\text{obs}}(\sigma_0) &\neq \rho_{\text{obs}}(\sigma_1), \\ \text{Obs}(\tau_0) &\neq \text{Obs}(\tau_1). \end{aligned} \quad (6)$$

Define the dependent tagged family

$$L_{\text{causal}}(V, I; K_{\text{obs}}) = \{(P, \ell, w) \mid w \text{ is causal at } (P, \ell)\}. \quad (7)$$

The artifact and frontier tags are part of the type. With fixed, decodable prefix encodings for artifacts, frontiers, and operation symbols, the family can be embedded in an ordinary language over one finite alphabet; the dependent notation keeps the carriers visible. A host descriptor is neither another element of L_V nor automatically a runtime word. It first becomes configuration, the accepted interpreter induces a schedule, and only a same-suffix witness enters L_{causal} .

The definition is observer- and slice-relative. Deleting an earlier input from the context after seeing the result would make the test circular. The `wm_circuit` interpreter case therefore declares its suffix from source semantics before comparison; its translated dump is retained for manual inspection rather than consumed by an automated slice checker. The separate `rac_single` carrier instead uses the hash-bound path-correlation check described in Section 5.7.

C. Behavioral quotient and report-relative residual language

Fix a deterministic discipline

$$D = (X_D, S_D, A_D, O_D, \delta_D, \lambda_D, s_D, \text{Obs}_D) \quad (8)$$

with an admissible concrete region $X_D \subseteq \Sigma_I$, state carrier S_D , operation alphabet A_D , output alphabet O_D , observer $\text{Obs}_D : O_D^* \rightarrow O_{\text{obs}}$, and same-domain partial functions

$$\delta_D : S_D \times A_D \rightarrow S_D, \quad \lambda_D : S_D \times A_D \rightarrow O_D. \quad (9)$$

Whenever D is used at (P, ℓ) , fix an injective operation encoding $\iota_{P,\ell} : A_D \rightarrow \Sigma_{\text{op}}(P)$ and extend it homomorphically. The pair $(s_D, \iota_{P,\ell})$ is *operationally adequate on X_D* when, for every $\sigma \in X_D$ and $a \in A_D$, concrete execution of $\iota_{P,\ell}(a)$ exists exactly when $\delta_D(s_D(\sigma), a)$ is defined; whenever $\sigma \xrightarrow{\iota_{P,\ell}(a)/o} \sigma'$, where $o \in O_D$ is the complete declared observation of that encoded operation, then $\sigma' \in X_D$ and

$$\lambda_D(s_D(\sigma), a) = o, \quad s_D(\sigma') = \delta_D(s_D(\sigma), a). \quad (10)$$

Thus, within X_D , the encoding is a semantic renaming rather than an arbitrary map: one projected state cannot hide different definedness, outputs, or successor projections. Every environment choice that affects a transition is fixed by D or included in S_D .

Let $\text{Out}_D(r, w)$ be the partial complete output word obtained by iterating (δ_D, λ_D) , write $\text{Def}_D(r, w)$ for $\text{Out}_D(r, w) \downarrow$, and take the continuation universe to be

$\mathcal{W}_D = A_D^*$; infeasible words are represented by undefined output. For $r, r' \in S_D$, define

$$\begin{aligned} r \sim_D r' &\iff \forall w \in \mathcal{W}_D. \\ &\left[\text{Def}_D(r, w) \iff \text{Def}_D(r', w) \right] \\ &\wedge \left[\text{Def}_D(r, w) \Rightarrow \text{Out}_D(r, w) = \text{Out}_D(r', w) \right]. \end{aligned} \quad (11)$$

Because $\mathcal{W}_D = A_D^*$ is closed under left-prefixing by an operation, $\sim_D$ is a right congruence: defined matching a -steps lead to equivalent successor states, since every successor continuation w is tested as aw . This follows the classical continuation-equivalence lineage of Nerode and Mealy machines [6], [7]; the partial-output setting and observer contract here are paper-specific. If $Q_D = S_D / \sim_D$ is finite, execution restricted to X_D induces a partial Mealy transducer over quotient states. Observational completeness and generalized strong preservation likewise ask whether an abstraction preserves a selected observation or specification language [8], [9]. Our test is narrower: it concerns cells actually computed for one accepted artifact at one frontier. It is a semantic quotient test, not a claim to introduce a new general completeness theory.

For a concrete set F , define the common enabled continuations

$$\mathcal{W}_D(F) = \{w \in \mathcal{W}_D \mid \forall \sigma \in F. \text{Out}_D(s_D(\sigma), w) \downarrow\}. \quad (12)$$

Proposition 1 (behavioral factorization on a context fiber).

Fix accepted P , frontier ℓ , discipline D , and observation contract K_{obs} . Let $\emptyset \neq F \subseteq \text{Reach}_I(P, \ell) \cap X_D$. Fix an injective operation encoding $\iota_{P, \ell} : A_D \rightarrow \Sigma_{\text{op}}(P)$ and its homomorphic extension to words. Assume

$$\iota_{P, \ell}(\mathcal{W}_D(F)) \subseteq W_{\text{run}}^\ell(P), \quad (13)$$

that K_{obs} is sound on F for every encoded word $\iota_{P, \ell}(w)$ with $w \in \mathcal{W}_D(F)$, and that

$$\forall w \in \mathcal{W}_D(F). \forall \sigma, \sigma' \in F. \text{ctx}_{\iota_{P, \ell}(w)}(\sigma) = \text{ctx}_{\iota_{P, \ell}(w)}(\sigma'). \quad (14)$$

Require observation compatibility: whenever encoded $w \in \mathcal{W}_D(F)$ executes from $\sigma \in F$ with concrete trace τ , the following right-hand side is defined and

$$\text{Obs}(\tau) = \text{Obs}_D(\text{Out}_D(s_D(\sigma), w)). \quad (15)$$

Define

$$\beta_D(\sigma) = [s_D(\sigma)]_D, \quad F_{a^\#} = F \cap \gamma_\ell(a^\#). \quad (16)$$

Finally, require the unique-cell condition

$$\forall \sigma \in F. \exists! a^\# \in \text{Report}_V(P, \ell). \sigma \in \gamma_\ell(a^\#), \quad (17)$$

and let $\pi_R : F \rightarrow \text{Report}_V(P, \ell)$ map each state to that unique computed cell. Then the report factors the future-observation quotient on F ,

$$\exists h : \pi_R(F) \rightarrow Q_D. \beta_D = h \circ \pi_R, \quad (18)$$

if and only if

$$\forall a^\# \in \pi_R(F). |\beta_D(F_{a^\#})| \leq 1. \quad (19)$$

Consequently, report non-factorization on F is equivalent to a unique computed cell containing two states from different $\sim_D$ classes.

Proof. If $\beta_D = h \circ \pi_R$, states with the same unique report label have the same quotient class, giving the cardinality bound. Conversely, the bound makes $h(a^\#)$ the unique element of $\beta_D(F_{a^\#})$ for every $a^\# \in \pi_R(F)$; this is well-defined and yields the factorization. Negating either equivalent condition gives the collision statement. $\square$

Write $\text{Adm}(P, \ell, D, F; K_{\text{obs}})$ when all of the following hold: $P \in L_V$; $(s_D, \iota_{P, \ell})$ is operationally adequate on X_D ; $\emptyset \neq F \subseteq \text{Reach}_I(P, \ell) \cap X_D$; the runtime-word inclusion, observer compatibility, sound observation contract, common-context condition, and unique-cell condition above all hold. The last condition makes the nonempty $F_{a^\#}$ a disjoint cover. Reports with overlapping labels require a separately declared membership-signature map and are outside this criterion.

Admissibility is an internal semantic typing condition, not a safeguard against post-selection. Every positive instantiation must therefore declare whether D , F , the observer, suffix, and report interface were fixed prospectively or selected retrospectively after the distinguishing execution was known. A prospective or report-general claim additionally requires selection provenance independent of the witnessed output, such as a predeclared protocol or an externally specified report contract. A retrospective tuple may satisfy Definition 2 exactly as declared, but that fact alone cannot characterize the recognizer's intended report, a broader execution domain, or a population of implementations.

Definition 2 (report-relative residual language). Fix an admissible tuple. A dependent tagged word $(P, \ell, D, F, a^\#, w)$ is *output-witnessed report-relative residual* when:

$$\begin{aligned} (P, \ell, \iota_{P, \ell}(w)) &\in L_{\text{causal}}(V, I; K_{\text{obs}}), \quad w \in \mathcal{W}_D(F), \\ \sigma_0, \sigma_1 &\in F \text{ are Definition 1 witnesses for } \iota_{P, \ell}(w), \\ \pi_R(\sigma_0) &= \pi_R(\sigma_1) = a^\#, \quad \beta_D(\sigma_0) \neq \beta_D(\sigma_1). \end{aligned} \quad (20)$$

$L_{\text{res}}^R(V, I, \text{Report}; K_{\text{obs}})$ is the union of these tagged words over tuples satisfying Adm . Operational adequacy and observer compatibility make different admitted concrete observations imply different quotient classes; the explicit inequality binds the definition to Proposition 1. The quotient also distinguishes definedness, so report non-factorization can arise without a jointly terminating output-distinguishing word. Accordingly, L_{res}^R is the output-witnessed subset of report-factorization failures, not a converse characterization of all failures. It is a frontier collision, not a claim that a complete whole-program report can never recover the final function.

Instantiating Definition 2 requires an extractor for actual computed cells, a declared concretization, a unique-cell partition on the chosen fiber (or a separately defined alternative report map), and the remaining admissibility premises. The retained `wm_circuit` Linux log does not provide these objects, so that interpreter carrier gives only a conditional L_{causal} witness under its declared service contract and does

not establish L_{res}^R . Section 5.7 instantiates those objects for a different, finite `rac_single` carrier.

D. Shape, defect, and policy

A defect-induced gap uses behavior that violates the declared recognizer, report, or runtime contract. A documented, safety-preserving witness in L_{res}^R for a relation the report intends to certify is evidence for a contract-shape gap, not by itself a shape proof: report conformance and granularity must rule out a faulty transformer, join, or extractor. A policy-level weird machine additionally needs linked actor control, a policy-excluded effect, and unintended interpretation. An ordinary stateful API can therefore implement a transducer without satisfying either classification.

III. BOUNDED STATE-MEDIATED CIRCUIT REALIZATION

A. From a distinction to a reusable gate

A causal word can expose one distinction without supporting repeated computation. Fix a deterministic gate discipline D , one canonical reset class $q_0 \in Q_D$ identified with its subset of S_D , and a nonempty admissible set $\text{Adm}_G \subseteq X_D$. A gate basis $(P_G, \text{reset}, G, \text{observe}, D, \text{Adm}_G)$ uses one accepted artifact P_G , a dispatcher G selecting a complete word $u_G(x) \in A_D^*$ for each $x \in \{0, 1\}^2$, and a partial readout $\text{observe} : O_D^* \rightarrow \{0, 1\}$. For this basis, $O_{\text{obs}} = \{0, 1\}$ and $\text{Obs}_D = \text{observe}$. Its predeclared operation encoding into $\Sigma_{\text{op}}(P_G)$ is operationally adequate on X_D . The basis satisfies:

E1 — causal basis and observation. There exist inputs x_0, x_1 , prefixes $p_0, p_1 \in A_D^*$, one common remaining suffix $w \in A_D^*$, reset-class states $\eta_0, \eta_1 \in \text{Adm}_G$, and one internal frontier ℓ_G such that

$$u_G(x_i) = p_i w, \quad s_D(\eta_i) \in q_0. \quad (21)$$

Executing each p_i reaches a state σ_i at ℓ_G ; the two σ_i are Definition 1 witnesses for the encoded common suffix w . If its traces are τ_i , then the complete gate readout is precisely that causal observation:

$$\text{observe}(\text{Out}_D(s_D(\eta_i), u_G(x_i))) = \text{Obs}(\tau_i), \quad i \in \{0, 1\}. \quad (22)$$

Accepted code uses this readout as the gate result that E4-D writes; it is not an unrelated probe.

E2 — uniform input control. The dispatcher G in P_G reads runtime bits $x \in \{0, 1\}^2$. For every $\sigma \in \text{Adm}_G$ with $s_D(\sigma) \in q_0$, $\text{Out}_D(s_D(\sigma), u_G(x))$ and its readout are defined and

$$\text{observe}(\text{Out}_D(s_D(\sigma), u_G(x))) = g(x) \quad (23)$$

for one fixed $g : \{0, 1\}^2 \rightarrow \{0, 1\}$. The experimenter is not an external oracle choosing a different constant program for each input.

E3 — reset. The reset word satisfies $\text{reset} \in A_D^*$. From every $\sigma \in \text{Adm}_G$, it is defined, preserves declared wire cells, and terminates in $\tau \in \text{Adm}_G$ with $s_D(\tau) \in q_0$. If two pre-states

in Adm_G agree on those wire cells and the fixed context and reset to τ_0, τ_1 , then for every $x \in \{0, 1\}^2$,

$$\text{Out}_D(s_D(\tau_0), u_G(x)) = \text{Out}_D(s_D(\tau_1), u_G(x)). \quad (24)$$

Without E1, a hidden difference has no program-visible effect. Without E2, the experimenter may supply the truth table. Without E3, the channel may be one-shot. A fourth condition below composes the gate without placing circuit semantics in an external oracle.

B. Descriptor domain

The artifact uses the bounded domain $D_{64,512}$. A descriptor is

$$d = (m, n, (s_i^0, s_i^1)_{0 \leq i < n}), \quad (25)$$

with $0 \leq m \leq 64$, $0 \leq n \leq 512$, and

$$0 \leq s_i^b < 2 + m + i \quad (26)$$

for each gate i and operand b . Write $m(d) = m$ and $n(d) = n$. Wire 0 is constant zero, wire 1 is constant one, primary inputs occupy wires 2 through $2 + m - 1$, and gate i writes canonical destination $2 + m + i$. Thus the maximum number of live canonical wires is $2 + 64 + 512 = 578$ and the highest valid wire index is 577.

For $x \in \{0, 1\}^{m(d)}$ and gate function g , $\text{Eval}_g(d, x)$ initializes the constants and input vector, then extends the wire vector in descriptor order:

$$\nu[2 + m + i] = g(\nu[s_i^0], \nu[s_i^1]). \quad (27)$$

Its result is the complete canonical vector $(\nu[0], \dots, \nu[1 + m(d) + n(d)])$, the same type later returned by WireObs_d .

The host encoding $\text{Enc}_U(d, x)$ writes a normalized descriptor, inputs, and control record into maps. Textual WMC1 is a host serialization. Neither WMC1 nor d is a BPF program accepted by V ; $\text{Sched}_U(d, x)$ is the operation schedule induced when accepted P_U reads the encoding.

For $c = \text{Enc}_U(d, x)$, let $\text{PhysRun}_U(c) = (s, k, \mu)$ contain final status, completed-iteration count, and physical map state. Define the descriptor-relative canonical observation

$$\text{WireObs}_d(\mu) = (\mu[0], \dots, \mu[1 + m(d) + n(d)]) \quad (28)$$

and the status-masked interface

$$\text{Run}_{U,d}(c) = \begin{cases} (s, \text{WireObs}_d(\mu)), & s = \text{OK}, \\ (s, \perp), & s \neq \text{OK}. \end{cases} \quad (29)$$

The host may project requested output wires only from an OK result. This semantic rule does not assert that stale physical cells are erased.

E4-D — bounded data-parametric interpretation. One fixed artifact P_U with a gate basis whose artifact component is the same P_U discharges E4-D for $D_{64,512}$ when:

- 1) for the fixed map definitions and recorded load environment, $P_U \in L_V$ independently of descriptor contents d and x ;

- 2) every valid $c = \text{Enc}_U(d, x)$ establishes a pre-callback state $\mu^{(0)}$ with $\mu^{(0)}[0] = 0$, $\mu^{(0)}[1] = 1$, and $\mu^{(0)}[2 + j] = x_j$ for $0 \leq j < m(d)$, executes exactly callback positions $0, \dots, n(d) - 1$ in order, and terminates with $\text{PhysRun}_U(c) = (\text{OK}, n(d), \mu)$;
- 3) one external critical section covers setup, invocation, and readback for every shared map in the footprint;
- 4) every iteration i validates canonical form and copies exactly its two earlier source wires; the complete concrete state σ_i immediately before reset belongs to Adm_G ; the iteration then resets and invokes the E1–E3 gate, writes the resulting g bit only to destination $2 + m(d) + i$ and declared audit cells, and preserves the descriptor and all earlier wires; and
- 5) no execution performs more than 512 iterations, while malformed core-ABI controls or descriptor configurations covered by the declared validator terminate with non-OK status, at most 512 completed iterations, and semantic result $(s, \perp)$.

These clauses are deliberately proof obligations. In particular, E4-D requires the gate result to be written and exact iteration/terminal behavior; merely traversing a descriptor is insufficient.

C. Realization theorem

Theorem 1 (bounded composition under explicit obligations). Assume the fixed P_U discharges E4-D under its declared deterministic and serialized environment, and the gate basis on that same P_U satisfies E1–E3 with function g . Then, for every $d \in D_{64,512}$ and $x \in \{0, 1\}^{m(d)}$,

$$\text{Run}_{U,d}(\text{Enc}_U(d, x)) = (\text{OK}, \text{Eval}_g(d, x)) \quad (30)$$

after exactly $n(d)$ iterations. If the recognition boundary is additionally safety-sound for Safe, these traces also satisfy Safe.

Proof sketch. E4-D initializes the constant/input prefix. At iteration i , the descriptor bound makes both sources earlier cells. E3 supplies q_0 , E2 identifies the readout with g from that class, and E4-D writes that same readout only to the canonical destination while preserving the established prefix. Prefix induction yields Eval_g after $n(d)$ iterations; the exact-schedule clause supplies terminal OK. E1 binds the bit used by E2 and E4-D to a causal same-suffix state distinction; without E1 the functional induction would show only an ordinary bounded g -evaluator, not a state-mediated one. Safety follows only from the separate soundness premise. $\square$

The theorem decomposes local gate, reset, scheduling, and frame obligations; it is not a claim that tests enumerate the descriptor domain. It also does not prove E4-A, in which a compiler emits a different accepted BPF object per circuit.

IV. EBPF CALIBRATION CASE

A. Execution boundary

The program $P_U = \text{wm_circuit}$ is one fixed eBPF artifact with section $\text{SEC}(\text{syscall})$. In the recorded environment the

in-kernel verifier accepts it and userspace executes it offline through $\text{bpf_prog_test_run_opts}()$; Linux documents both userspace program execution and the syscall section/program-type mapping [10], [11]. No live hook is involved. The experiment is privileged and local, and its acceptance and resource facts remain specific to the recorded program type, kernel, and architecture.

The carrier boundary is:

The gate map $G0$ is a dedicated BPF_MAP_TYPE_HASH map with $\text{max_entries} = 2$. It is non-LRU and retains the default preallocation; Linux documents the entry bound, default preallocation, update flags, and success/negative-error convention for this map type [12]. The discipline requires successful reset and one external critical section over setup, invocation, and readback for every map in the footprint. The serial harness satisfies the no-interleaving use pattern, but the eBPF program implements no concurrency lock.

B. Saturating-rank NAND

The gate uses pairwise-distinct keys: sentinel S and input keys A and B . Reset deletes all three and inserts S , establishing occupancy one. For the first input, zero updates S and one inserts fresh A ; for the second, zero updates S and one inserts fresh B . Proposition 2 states the abstract occupancy law. For the Linux instance, existing-key success, below-capacity fresh-key success, and at-capacity fresh-key failure are explicit premises restricted to valid arguments, default preallocation, successful reset, no interference, and the recorded Linux 6.17 environment. Reference [12] supplies the map interface and return convention; retained raw returns and mechanism controls instantiate the law for this object. The gate observes only the second update's success predicate, not a portable error number.

The four executions are:

Proposition 2 (saturating-rank NAND). Let rank mean occupied-name count in a resource of capacity $k \geq 2$. Assume reset establishes rank $k - 1$, S is among those occupied names, and pairwise-distinct A and B are fresh. Existing-name update succeeds without changing rank; fresh-name update succeeds and increments rank below k , and fails without changing rank at k . Dispatch zero to S , the first one to A , and the second one to B . The second update's success predicate is $\text{NAND}(a, b)$.

Proof. When $a = 0$, the first update preserves rank $k - 1$, so either second update succeeds. When $a = 1$, the first update raises the rank to k ; the second update succeeds for $b = 0$ because S already exists and fails for $b = 1$ because B is fresh. The outputs are therefore 1, 1, 1, 0. $\square$

NAND is functionally complete because $\neg x = \text{NAND}(x, x)$ and $x \wedge y = \neg \text{NAND}(x, y)$; the 512-gate descriptor bound limits circuit size, not the Boolean basis.

For the concrete Definition 1 witness, take $P = \text{wm_circuit}$ and let ℓ be the point in circuit_step_cb immediately before the second map update, after the first input-conditioned operation. Let $R_{G0}(\sigma)$ be the complete helper-relevant dynamic

TABLE II
CARRIER BOUNDARY IN THE EBPF CALIBRATION CASE.

Host data and validation	Recognition unit	Post-acceptance interpretation
textual WMC1 $\rightarrow$ normalized map configuration $\text{Enc}_U(d, x)$	V accepts the fixed P_U , not WMC1 or d	P_U reads maps $\rightarrow \text{Sched}_U(d, x) \rightarrow$ helper returns and wire observations

TABLE III
SATURATING-RANK NAND EXECUTIONS.

a	b	Operation word after reset	State before second operation	Second result	Output [$ret = 0$]
0	0	$updS; updS$	$\{S\}$	success	1
0	1	$updS; insB$	$\{S\}$	success	1
1	0	$insA; updS$	$\{S, A\}$	success	1
1	1	$insA; insB$	$\{S, A\}$	failure	0

state of the dedicated map G_0 —including key occupancy, buckets, preallocated element/free-list metadata, and any other map-local state read by the update helper—and set $\rho_{\text{obs}}(\sigma) = R_{G_0}(\sigma)$. The occupied-key set $K_{G_0}(\sigma)$ is only a derived projection of $R_{G_0}(\sigma)$. Inputs $(0, 1)$ and $(1, 1)$ reach states σ_0 and σ_1 at ℓ with $K_{G_0}(\sigma_0) = \{S\}$ and $K_{G_0}(\sigma_1) = \{S, A\}$, hence $R_{G_0}(\sigma_0) \neq R_{G_0}(\sigma_1)$.

Take the common remaining suffix w to be the fresh- B update followed by the uniform capture and store of its success bit, and define $\text{Obs}(\tau) = [ret(\tau) = 0]$. The slice and ctx_w include the common program point, key B , value, G_0 identity and static attributes (map type, capacity, default preallocation, and BPF_ANY mode), relevant control and wire values, and every suffix-read component outside R_{G_0} . The contract's Env fixes the kernel, object, map instances, schedule, and absence of interference. Under the stated map-update contract, the two suffixes terminate with observations 1 and 0. The earlier input is not read by the suffix, and its persistent suffix-relevant effect is contained in the selected complete G_0 dynamic state, so the non-selected contexts agree. Thus w is a conditional Definition 1 witness under this declared concrete-service contract. This is stronger than selecting occupancy alone and does not claim that the retained traces expose all internal kernel fields.

Under the gate discipline, take $s_{D_G}(\sigma) = (\text{phase}(\sigma), K(\sigma))$ on the invariant region X_{D_G} fixed by valid arguments, the dedicated preallocated map contract, successful reset, serialization, and no interference. Adequacy on this region is an explicit service-contract premise, not a machine-checked model of every kernel field. Phase ranges over the reset/sentinel/first/second-operation stages and $K \subseteq \{S, A, B\}$ with $|K| \leq 2$. The carrier, and hence its future-observation quotient, is finite. For E1, the two complete gate words for $(0, 1)$ and $(1, 1)$ have prefixes $updS$ and $insA$ and the common remaining suffix just defined. The complete reset-plus-gate word need not be causal because reset erases incoming differences; the witness is tagged by the internal frontier after the first operation.

For the basis take $P_G = P_U$, $\text{Adm}_G = X_{D_G}$, reset as delete S, A, B then insert S, G as the $S/A/B$ selector, and $\text{observe} = [\text{second return} = 0]$. The witness above discharges

E1; Proposition 2 and the reset premises discharge E2–E3.

This mechanism adds no extensional Boolean expressiveness to eBPF. It is a calibration case showing that a documented stateful service can supply a reusable post-acceptance gate; it does not by itself establish report omission.

C. Fixed interpreter and map ABI

In addition to G_0 , $wm_circuit$ uses *TAPE* and four interpreter maps:

- *TAPE* records build variant, capacity, selected raw return, and error count;
- *CIRCUIT* stores normalized records ($op, src0, src1, dst$);
- *WIRES* stores constants, primary inputs, and canonical SSA-style gate outputs;
- *VM_CONTROL* supplies the ABI version and declared counts;
- *VM_TRACE* stores per-gate validity, output, and, for state-mediated variants, the raw second-helper return.

The host parser checks and normalizes WMC1, writes map cells, and retains the requested output list. The list is projected only after $\text{status} = \text{OK}$ and is never read by BPF. The source-level mask is guarded by host control flow; the negative suite tests rejection status and execution counts, not an injected runtime masking path. Inside the kernel, a bounded loop handles at most 512 descriptors. Each iteration copies descriptor and source values before helper calls, resets G_0 , invokes the gate, and updates the canonical destination by key, avoiding retention of map-value pointers across those calls.

Canonical destinations prevent source/destination aliases and forward references. Covered malformed versions, counts, opcodes, destinations, or sources receive non-OK status. For valid descriptors, the topological restriction makes every source earlier. External whole-transaction serialization prevents concurrent mixed configurations; successive serial invocations overwrite shared cells, and stale physical cells may remain.

The verifier's acceptance unit is the fixed interpreter artifact rather than each descriptor, although the harness may reload the same object for different datasets. Changing map

configuration changes the interpreted circuit. This is the E4-D carrier distinction, not evidence for a compiler that emits a new accepted BPF object per circuit.

D. Source-level invariant and evidence boundary

Lemma 1 (source-level interpreter prefix invariant). Assume the update laws of Proposition 2, successful reset and required map/helper calls, the E4-D serialization environment, valid $\text{Enc}_U(d, x)$, and that the captured translated object preserves the manually inspected source control/data dependencies. For every integer t with $0 \leq t \leq n(d)$, after successful completion of the ordered callback indices $0, \dots, t-1$, every canonical wire below $2 + m(d) + t$ equals the corresponding prefix of $\text{Eval}_{\text{NAND}}(d, x)$.

Proof sketch. The outer program initializes constants and preserves host-written inputs. A callback validates the current opcode, destination, and earlier sources, then copies the descriptor and source bits. Proposition 2 gives NAND after reset; success writes only the canonical destination and increments the completed count. The outer program requests $n(d)$ iterations through *bpf_loop*; under the helper contract, callback return zero continues, return one stops, and the helper reports the number of iterations performed [13]. The interpreter accepts only when both that return and its own completed count equal $n(d)$. Induction on t gives the prefix claim. This is a source-level argument supported by retained translated dumps for manual inspection, not a machine-checked eBPF-semantics proof. $\square$

The preserved object, verifier metadata, and loaded-program tag support E4-D’s fixed-acceptance unit. Source and translated dumps support initialization, ordered *bpf_loop* use, validation, reset, one-destination writes, and bounds; audits exercise outputs and negative status/count behavior. Serialization remains external because the program has no lock. Thus A is recorded, C has a conditional witness under the declared map-service/no-interference contract, and P is supported under additional source-to-object, frame, and serialization premises. For this *wm_circuit* carrier, nodes R and W are absent: no interpreter-specific Linux report-cell extractor or deployment policy is supplied.

V. EVALUATION

A. Recorded environment and primary run

The primary evidence is the run bundle `results/interpreter/interpreter-final-20260711-02/`. It records Ubuntu 24.04, Linux 6.17.0-35-generic on aarch64, and preserves four BPF variants, captured loaded-program metadata, verifier logs, descriptors, JSONL outputs, an author-run semantic audit, a then-current selected source/manuscript snapshot, and a self-issued SHA-256 manifest. The final manuscript postdates that run and is not claimed to be covered by its manifest. The bundle is author-generated and separately author-audited, not a third-party reproduction.

The dataset contains exactly

$$\begin{aligned} 38,533 &= 26,488 \text{ per-gate records} \\ &+ 12,037 \text{ successful run records} \\ &+ 8 \text{ negative-control records.} \end{aligned} \quad (31)$$

These are heterogeneous evidence rows, not “38,533 tests.” In particular, the explicit arithmetic baseline’s gate records do not contain state-mediated helper-return traces.

B. Coverage

Named and random circuits exhaust their own finite inputs; the joint 64-input boundary uses only all-zero and all-one vectors, not 2^{64} inputs. The corpus is regression evidence, not an enumeration of $D_{64,512}$. Serial reuse detects common reset/stale-state regressions but is neither a concurrent stress test nor proof of the external critical-section premise.

C. Separate semantic audit and integrity check

The author-run auditor does not rely solely on the runner’s pass flag: it also reconstructs named and boundary descriptors, regenerates the seeded 100-DAG corpus, independently evaluates complete wire vectors, checks destinations, boundary gates and eight malformed cases, and matches runtime tags to loaded-program metadata. It reports success. A subsequently written self-issued SHA-256 manifest also verifies; it detects drift but is not a signature, timestamp, attestation, or independent reproduction.

D. Mechanism attribution

Three variants separate the capacity mechanism from the truth table and artifact acceptance.

First, increasing capacity from 2 to 64 removes saturation in the tested schedule; all 166 named-corpus control gate outputs become 1. Second, forcing the second one-bit input to update S removes the fresh- B insertion; again all 166 outputs become 1. Third, an arithmetic baseline computes NAND without the capacity mechanism and produces the expected named-circuit results. All four variants—the state-mediated gate, two controls, and baseline—load and execute in the recorded environment.

The controls support mechanism attribution: saturation and the second fresh name are both necessary for the gate’s zero output. They do not show equivalent verifier reports. The baseline confirms that the mechanism gains no new Boolean function unavailable to ordinary bytecode.

E. Calibration result

Linux’s verifier documentation states its objective in terms of determining program safety and validating paths, arguments, and memory access [14]; it does not specify a complete functional certificate for map-mediated computation. The retained *wm_circuit* evidence has no report extractor at its interpreter frontier, so we neither test nor infer how precisely a verifier cell tracks that program’s helper return. This carrier is a LangSec calibration of the boundary between recognized safety and interpreted functionality: it gives a conditional C

TABLE IV
RECORDED EVALUATION COVERAGE.

Dataset and input coverage	Successful runs	Per-gate records
9 named circuits, exhaustive per circuit	39	166
100 random DAGs, exhaustive per DAG (≤ 6 inputs, ≤ 24 gates; seed 3235823838)	1,876	23,776
512-gate deep and 64-input/512-gate/578-wire joint boundaries	4	2,048
zero-gate descriptor; 10,000 serial NAND/adder/mux invocations	10,001	0
capacity-64, forced-sentinel, and explicit-logic controls	117	498
8 malformed ABI/count/op/destination/reference cases	—	—

witness under the declared service contract and supports P under additional premises, but it is not evidence of unsound acceptance and its interpreter-specific R node remains unestablished. Section 5.7 evaluates R on a separate stock-Linux carrier with a dedicated accepted object and an explicitly declared operational prune-report contract.

F. Fixed auxiliary executable report instance

To exercise the report-relative criterion without relabeling Linux verifier artifacts, we embed the auxiliary R instance in the full model carrier of Section 7.1 and fix

$$M_{linux_r_aux_v1} = (V_{linux_r}, I_{hash}, \text{Report}_{aux}, K_{obs}, P_{aux}, \ell, D, F, \emptyset, \emptyset, \emptyset, \emptyset). \quad (32)$$

P_{aux} is accepted by the custom report-producing recognizer V_{linux_r} ; it is not a BPF object accepted by the stock Linux verifier. I_{hash} is a finite, deterministic, serialized, no-interference semantics restricted to the non-evicting HASH-map update cases in the fixed program. Here Report_{aux} means only `report.json["report_cells"]`; `derivation.json` records `worklist/computation` provenance and is not part of the report-label interface. The last four components are the typed empty actor set, effect set, drive relation, and permitted-effect set; they embed the R instance in the common carrier and make no W claim.

Executable certificate. $R(M_{linux_r_aux_v1})$ holds (artifact status: established).

Certificate argument. The recognizer accepts P_{aux} . Exhausting $a \in \{0,1\}$ with $b = 1$ in the singleton serialized/no-interference environment gives $F = \text{Reach}_{I_{hash}}(P_{aux}, \ell) = \{\text{frontier:S}, \text{frontier:AS}\} \subseteq X_D$. Here X_D additionally contains the corresponding two terminal states. Let $a_{obs} \in A_D$ be the discipline action named `update-suffix-and-observe`, write c_{obs} for the exact encoded runtime-operation symbol `bpf_map_update_elem(G0, suffix_key, one, BPF_ANY); observe(ret==0)`, and set $\iota_{P_{aux}, \ell}(a_{obs}) = c_{obs}$; this singleton encoding is injective. Concrete execution of $\iota_{P_{aux}, \ell}(a_{obs})$ is defined on the two frontier states, undefined on the terminal states, and agrees with D on definedness, output, and successor. Thus the checker establishes operational adequacy on all of X_D . The observation contract fixes ρ_{obs} as the occupied-key set, Obs as the ordered success-bit word, Slice as the program-phase/service-context pair, and Env as that singleton environment; in particular, $\rho_{obs}(\text{frontier:S}) = \{S\} \neq \{S, A\} = \rho_{obs}(\text{frontier:AS})$. For every ordered pair in

F^2 and every word in $\mathcal{W}_D(F) = \{\varepsilon, a_{obs}\}$, the checker establishes the remaining runtime-word, common-context, observer-compatibility, and K_{obs} -soundness obligations. These discharge every admissibility item except unique-cell coverage.

Report_{aux} emits one unique cell $a^\#$ that covers both frontier states, completing $\text{Adm}(P_{aux}, \ell, D, F; K_{obs})$. The exact finite future-observation quotient assigns them different classes, so $|\beta_D(F_{a^\#})| = 2$; moreover a_{obs} yields observations 1 and 0. Hence the same suffix first witnesses Definition 1 and the tagged tuple $(P_{aux}, \ell, D, F, a^\#, a_{obs})$ satisfies Definition 2. Therefore $R(M_{linux_r_aux_v1})$ holds. The evidence also records 21 domain/action *return-class containment* checks plus two report-cell *successor-containment* checks, all with zero violations; the former are not 21 full post-state checks. Each of the four negative controls—exact occupancy tracking, capacity 64, forced sentinel, and an unobserved return—removes the R witness. $\square$

On the archived bundle, a separately implemented, author-run checker reconstructs reachability, unique-cell coverage, the quotient, and the factorization verdict without importing the model implementation, and emits the established auxiliary-R verdict. Separately, one regression test builds the same deterministic model bundle twice and byte-compares its five formal JSON files; this is a determinism test, not experimental evidence. The certificate is executable finite-model evidence, not a machine-checked proof. The retained kernel calibration has four oracle rows for the two assignments (0, 1) and (1, 1); it calibrates those two restricted service outcomes only. It proves no refinement or bisimulation between I_{hash} and Linux and extracts no stock-verifier cell. The auxiliary certificate therefore remains independent of the stock-Linux certificate below. Its bundle is `results/linux_r/linux-r-v1/`.

G. Post-hoc stock-Linux trace certificate

This subsection analyzes a completed local stock-Linux experiment as a retrospective trace certification. The kernel capture is primary evidence that one prune event and two runtime outcomes occurred on the frozen tuple. The integrated and frozen-bundle checkers validate the declared evidence relationships offline; their regression suites are not substitutes for the capture, and neither layer converts the author-declared operational prune-report into a documented Linux functional-report contract.

To apply Definition 2 without reusing the auxiliary seman-

tics, we fix the hash-bound, evidence-restricted carrier

$$M_K = (V_K, I_K, \text{Report}_K^{\text{prune}}, K_{\text{obs}}^K, P_R, \ell_K, D_K, F_K, \emptyset, \emptyset, \emptyset, \emptyset). \quad (33)$$

P_R is the fixed verifier-accepted XDP object `rac_single`, not the interpreter $P_U = \text{wm_circuit}$. V_K is the recorded stock Linux 6.17.0-35-generic verifier at exact-comparison level 0, and $\ell_K = \text{pre}(41)$ is the translated caller-side frontier before the common `shared_suffix` call. For this retrospective proposition, I_K is a manuscript-defined finite, phase-tagged, serialized relation constructed after the capture by restricting and phase-tagging the bundle's two recorded executions. It is neither the bundle adapter's self-loop transition system nor the set of all Linux executions. Its declared input/environment domain contains exactly the two captured selector histories, so the following reachability equality is a definition of the restricted analysis carrier, not an empirical completeness claim about all states that Linux can reach:

$$\text{Reach}_{I_K}(P_R, \ell_K) = F_K = \{\sigma_0, \sigma_1\}. \quad (34)$$

Let $X_{D_K} = \{\sigma_0, \sigma_1, \sigma_0^+, \sigma_1^+\}$, where these symbols denote constructed, phase-tagged concretization witnesses for the two prefix cases and their post-suffix records. They are not direct snapshots of complete runtime registers, stack, and helper-internal map state. The two `runtime.json` final-state records bind only the derived G0 key-set projections $\{S, B\}$ and $\{S, A\}$ and their program-level outcomes; the verifier capture separately binds the selected State V2 records. Let a_B be the abstract insert- B action. Define a macro-symbol $c_B \in \Sigma_{\text{op}}(P_R)$ whose declared concrete interpretation executes the exact remaining translated suffix, and set $\iota_{P_R, \ell_K}(a_B) = c_B$. Fix the one-step discipline

$$\begin{aligned} S_{D_K} &= \{p_0, p_1, q_0^{\text{term}}, q_1^{\text{term}}\}, \\ A_{D_K} &= \{a_B\}, \quad O_{D_K} = \{0, 1\}, \\ O_{\text{obs}}^K &= \{\varepsilon, 0, 1\}, \\ s_{D_K}(\sigma_i) &= p_i, \quad s_{D_K}(\sigma_i^+) = q_i^{\text{term}}, \quad \delta_{D_K}(p_i, a_B) = q_i^{\text{term}}, \\ \lambda_{D_K}(p_0, a_B) &= 1, \quad \lambda_{D_K}(p_1, a_B) = 0. \end{aligned} \quad (35)$$

At the concrete level, the declared relation contains exactly

$$\sigma_i \xrightarrow{c_B/b_i} I_K \sigma_i^+, \quad (b_0, b_1) = (1, 0), \quad (36)$$

for this action and has no c_B -transition from σ_0^+ or σ_1^+ . These transitions are the manuscript wrapper; the bundle's `runtime.json` records bind their key-set projections and program-level outcomes, not complete concrete endpoints. There are no outgoing actions from $q_0^{\text{term}}, q_1^{\text{term}}$. Both observers return bit words in O_{obs}^K : explicitly, $\text{Obs}_{D_K}(\varepsilon) = \varepsilon$ and $\text{Obs}_{D_K}(b_i) = b_i$; on the corresponding declared concrete side, Obs maps the empty trace to ε and each recorded c_B trace to b_i . Thus observation compatibility covers both words in $\mathcal{W}_{D_K}(F_K) = \{\varepsilon, a_B\}$. The phase tags and displayed transitions make operational adequacy and runtime-word inclusion true by construction for this restricted relation. They do not independently validate a larger Linux transition semantics.

The stock-specific contract K_{obs}^K sets $\rho_{\text{obs}}^K(\sigma) = R_{G0}(\sigma)$, the complete helper-relevant dynamic G0 state defined in Section 4.2, and observes the program-level success-bit word. The recorded key set is only a derived projection K_{G0} : $K_{G0}(\sigma_0) = \{S\}$ and $K_{G0}(\sigma_1) = \{S, A\}$, so $R_{G0}(\sigma_0) \neq R_{G0}(\sigma_1)$ even though the records do not expose every internal field. The context fixes the caller phase and suffix, kernel/object/program and map identity, key/value constants, map type/capacity/flags, serialization, the single-artifact condition, and every suffix-read component outside R_{G0} . The checker compares the seven declared normalized runtime context fields. Separately, a manuscript read-set review covers the caller call/return path at translated PCs 41–44 and the `shared_suffix` callee at PCs 107–122. The callee overwrites its key/value working registers and stack slots; outside the helper it otherwise reads only fixed constants and the fixed G0 map identity, while the caller merely invokes the callee and returns its result. Map-local buckets, element/free-list metadata, and any other helper-read G0 state belong to R_{G0} ; the singleton hash-bound `Env` fixes the kernel, object, exact map instance, service/allocation choices, schedule, and absence of interference. The two records therefore have the same non-selected suffix-read context. Since the selected-state projections already differ, the contract's equal- R_{G0} premise on F_K relates only an identical state to itself; determinism of the declared one-step service relation gives soundness. The recorded success bits equal Obs_{D_K} , giving observer compatibility. Thus the runtime-word, common-context, operational-adequacy, observation-compatibility, and soundness premises are discharged on this explicitly finite carrier, not on general Linux execution. The read-set argument and the typing of $I_K, D_K, K_{\text{obs}}^K$ are manuscript-reviewed obligations, not claims about what the checker machine-verifies.

$\text{Report}_K^{\text{prune}}$ is an explicitly author-declared *operational prune-report* selected to study the captured event: it assigns an admitted prefix witness to the retained verifier-state representative of a successful `states_equal` comparison that actually causes `is_state_visited` to prune the current state. The directed prune edge is the membership relation; we do not reinterpret `states_equal` as a symmetric mathematical equivalence over complete concrete states. Linux documents the verifier as a safety mechanism, not this interface as a complete functional report over runtime map occupancy. The report is therefore an analysis projection extracted from concrete prune events, not a Linux-specified certificate whose intended semantics the experiment has falsified.

The frozen manifest binds the kernel release, BTF and configuration hashes, object SHA-256, loaded program id/tag/pin, and translated-bytecode SHA-256. The selected `fexit` event records both `states_equal_success=true` and `is_state_visited_prune=true` at instruction 41. Its retained and current histories are distinct and pass through the translated branch calls for $a = 1$ and $a = 0$, respectively, before reaching the same frontier. Their state hashes also differ (516c47f044cc3fc3 versus a66d912d58c91de4); membership follows the observed directed prune edge, not

hash equality. A reviewed path-correlation checker binds both histories to that frontier and to the same remaining translated suffix; the capture-completion record reports zero lost events and zero parse errors.

Normalized membership checks associate the constructed witness σ_0 with the current captured verifier state and σ_1 with the retained captured verifier state; they do not capture complete runtime frontier states. The runtime records bind the witnesses' derived G0 key-set projections to $\{S\}$ and $\{S, A\}$. Within the restricted exact-0 State V2 shape and the declared operational report, the shape and unique-cell checkers establish

$$\gamma_{\ell_K}^K(\text{retained:516c47f044cc3fc3}) \cap F_K = \{\sigma_0, \sigma_1\}. \quad (37)$$

Neither state belongs to any other extracted report cell on F_K . Hence

$$\pi_R(\sigma_0) = \pi_R(\sigma_1) = \text{retained:516c47f044cc3fc3}. \quad (38)$$

Under the jointly defined word a_B , the recorded *program-level success bits* are 1 from σ_0 and 0 from σ_1 ; they are not raw `bpf_map_update_elem` returns (0 on success and negative on failure). The unequal outputs imply $p_0 \not\sim_{D_K} p_1$ directly from the definition of $\sim_{D_K}$, so $\beta_{D_K}(\sigma_0) = Q_0 \neq Q_1 = \beta_{D_K}(\sigma_1)$ while the operational report map assigns both concrete states to the same cell. Acceptance and identity, reachability and path, direct membership, session completeness, unique-cell, quotient, factorization, and hash gates all pass. Together with the explicitly typed one-step construction above, these facts establish $\text{Adm}(P_R, \ell_K, D_K, F_K; K_{\text{obs}}^K)$ and satisfy Definition 2. Therefore

$$R(M_K) \quad (39)$$

holds for the declared finite frozen carrier. We report this as a *retrospective, trace-local, operational-prune-relative certificate*. The integrated checker emits `STOCK_LINUX_R_ESTABLISHED_FOR_FROZEN_TUPLE`, but that identifier denotes passage of the hash-bound gates encoded for this tuple; it is not a claim that Linux exposes or violates a documented complete functional report. The typing of $I_K, D_K, K_{\text{obs}}^K$ and the soundness argument above are the manuscript-level formal wrapper for the one-step records, not a mechanically checked theorem over the full Linux execution space. The bundled adapter presents each prefix case as self-looping when recomputing its stable quotient. Because the first a_B output already differs, its refinement splits the states at depth one; replacing those self-loops by the terminal transitions above preserves that split and the independently extracted operational report map. We do not identify the two quotient models. The required β -inequality and output-witness clause are proved by a_B ; no longer-word result is imported.

The certificate is carrier-, tuple-, observer-, suffix-, and report-relative. Because those choices were finalized after the distinguishing capture was available, it supports an existence statement for the declared retrospective tuple but not an independently precommitted or population-level Linux claim. It does not generalize I_K to uncaptured inputs, continuations,

or reachable states; generalize to other kernels, configurations, BTFs, compiler outputs, or eBPF objects; characterize general `states_equal` semantics; prove verifier unsoundness, a vulnerability, P, W, or a weird machine; or show that map occupancy belongs to a Linux-specified complete functional report. The concretization argument is restricted to constructed witnesses compatible with the captured State V2 records and locally future-observable fields. Because $P_R \neq P_U$, the certificate cannot be combined with the interpreter's P certificate to satisfy Link. The frozen evidence is under `residuality-auditor/stock-linux-r-proof/`.

H. Threats to validity

The experiment uses one kernel build and architecture. The argument relies on a dedicated preallocated non-LRU map, successful reset, distinct keys, the stated update laws, and whole-map-set mutual exclusion. Gate-emitting datasets preserve raw returns; the 10,000-run stress file records run/status/output evidence without per-gate raw rows. The proof uses only success versus failure and promises no portable error number.

Finite testing does not prove all $D_{64,512}$. Theorem 1 instead depends on the stated source-level initialization, validation, ordered iteration, gate, and frame obligations; the corpus seeks counterexamples. Translated dumps and verifier logs are retained for manual inspection and manifest hashing, not consumed by an automated data-flow checker. Status masking has a source-order guard, while the negative suite tests rejection/status rather than an injected masking path. There is no machine-checked eBPF-semantics proof, and production-verifier safety soundness is assumed only for the optional Safe conclusion.

The stock-Linux experiment is author-generated and author-reviewed on one frozen kernel/object tuple; it has not been independently reproduced. The operational report, two-history carrier, observer, and suffix were finalized retrospectively, so the result is vulnerable to selection effects even though the internal checks pass. The bundle binds BTF, configuration, and source-map metadata but does not preserve the target `verifier.c` function bodies; its restricted shape lemma is not a source-level theorem about general Linux `states_equal` behavior. The raw capture-stage analyzer intentionally withholds an R verdict until the bytecode frontier, path correspondence, report contract, and concretization relation are reviewed; the hash-bound integrated checker consumes those later certificates and emits its tuple-specific evidence verdict. The formal R conclusion additionally uses Section 5.7's finite typed wrapper. Treating the raw event, a state hash, or similar verifier-log text alone as R would not satisfy Definition 2.

VI. RELATED WORK

Language-theoretic security supplies the recognition/interpretation framing [1]–[4]; object-centric tracing reconstructs state-dependent low-level operation languages [15]. Weird-machine work studies unintended computation

and exploitability [16], [17], insecure compilation makes the policy boundary explicit [18], and proof-carrying-code work identifies shadow execution outside a proof model [19]. Valid-input computation also appears in well-formed RPM metadata, while vulnerability-flow types derive abstract weird machines [20], [21]. Unlike work that demonstrates rich behavior, our criterion asks whether a declared computed report actually factors a future-observation quotient at one accepted-artifact frontier.

Abstract interpretation supplies concrete/abstract, soundness, and completeness vocabulary [5], [22]; our criterion is limited to uniquely assigned cells actually computed at one frontier. MOAT isolates post-acceptance BPF effects [23], while mismorphism compares interpretations [24]. For eBPF, range-analysis verification and state embedding target verifier-logic soundness/coverage errors [25], [26]. VEP’s “programmability” means reducing verification-toolchain restrictions, while Rex addresses rejection-side language-verifier mismatch by replacing a separate static verifier with language-based safety and an extralingual runtime [27], [28]; P here instead means one fixed stock-verifier-accepted artifact interpreting a bounded family. DRACO checks functional specifications after verifier acceptance, and BPFVERIFY translates eBPF bytecode to bit- and memory-precise Horn clauses for functional verification [29], [30].

Recent preprints further separate the boundary: Heimdall treats higher-level defects surviving compilation and acceptance; Yaksha-Prashna extracts third-party bytecode behavior; bpfix localizes proof loss in rejected programs [31]–[33]. Trust-boundary semantic-gap work studies insufficient assertions after correctly implemented syntactic acceptance [34]. They do not jointly distinguish the obligations in our claim graph. Conversely, our `wm_circuit` interpreter carrier does not establish R or W; the explicitly finite frozen stock-Linux carrier and the auxiliary tuple establish R only on their own carriers and do not supply P or W. These four 2026 works are cited as preprints, not peer-reviewed publications.

VII. LIMITATIONS, IMPLICATIONS, AND OUTLOOK

A. The claim graph has strict separations

To put all five nodes on one carrier, fix a model

$$M = (V, I, \text{Report}, K_{\text{obs}}, P_*, \ell_*, D_R, F_R, \mathcal{A}, \mathcal{E}, \text{Drive}, \text{Pol}), \quad (40)$$

enriched with named intended-semantics, safety, report-abstraction, and granularity contracts against which the evidence predicates below are evaluated, where $\mathcal{A}$ and $\mathcal{E}$ are actor and effect sets, $\text{Drive} \subseteq \mathcal{A} \times \mathcal{E}$ records effects actors can drive through the designated P_* , and $\text{Pol} \subseteq \mathcal{E}$ is the permitted-effect set. Define $A(M)$ by $P_* \in L_V$; $C(M)$ by $\exists \ell, \exists w. (P_*, \ell, w) \in L_{\text{causal}}(V, I; K_{\text{obs}})$; $P(M)$ when P_* is one fixed accepted bounded interpreter with a same-artifact basis $(P_*, \text{reset}, G, \text{observe}, D_G, \text{Adm}_G)$ satisfying E1–E3, P_* discharges E4–D using that basis, and g together with constants $\{0, 1\}$ is functionally complete for Boolean circuits; $R(M)$ when $\text{Adm}(P_*, \ell_*, D_R, F_R; K_{\text{obs}})$

holds and some $(P_*, \ell_*, D_R, F_R, a^\#, w)$ is output-witnessed report-relative residual on exactly that tuple; and $W(M)$ when $\exists a \in \mathcal{A}, \exists e \in \mathcal{E}. (a, e) \in \text{Drive} \wedge e \notin \text{Pol}$.

For classification, let $\text{Link}(M)$ require the R witness (when used) to lie in the gate/interpreter execution family establishing P and the W effect to arise by an actor driving that same encoded computation; unrelated co-resident witnesses do not qualify. Let $\text{Unint}(M)$ mean that this programmable interpretation/effect is outside the boundary’s intended semantics. Define

$$\text{WM}_{\text{policy}}(M) = P(M) \wedge W(M) \wedge \text{Link}(M) \wedge \text{Unint}(M). \quad (41)$$

Let $\text{Doc}(M)$ mean that this linked behavior uses documented semantics and preserves the declared safety contract, $\text{Conf}(M)$ mean that the report implementation conforms to its specified abstraction, and $\text{Gran}(M)$ mean that its R collision is forced by the declared report granularity. Define the narrower class by

$$\text{WM}_{\text{shape}}(M) = \text{WM}_{\text{policy}}(M) \wedge R(M) \wedge \text{Doc}(M) \wedge \text{Conf}(M) \wedge \text{Gran}(M). \quad (42)$$

Proposition 3 (non-implications among claim nodes). Over this model class, the following implications are invalid:

$$\begin{aligned} A \not\Rightarrow C, \quad C \not\Rightarrow P, \quad C \not\Rightarrow R, \quad P \not\Rightarrow R, \\ R \not\Rightarrow P, \quad P \not\Rightarrow W, \quad R \not\Rightarrow W. \end{aligned} \quad (43)$$

The definitions validate $R \Rightarrow C \Rightarrow A$ and $P \Rightarrow C \Rightarrow A$: Definition 2 contains a Definition 1 witness, and E1 uses the E4–D artifact.

Proof by finite countermodels. For $A \not\Rightarrow C$, accept only *skip*. For $C \not\Rightarrow P$ and $R \not\Rightarrow P$, use states $\{0, 1, \perp\}$, a one-shot bit read, and one report cell. Exact future-class cells give $C \not\Rightarrow R$; an E1–E4–D NAND interpreter with that exact partition gives $P \not\Rightarrow R$. Setting $\text{Pol} = \mathcal{E}$ gives $P \not\Rightarrow W$ and, in the one-shot model, $R \not\Rightarrow W$. Finally, an actor-driven, unintended, policy-excluded effect from the same encoded P family with an exact report partition gives $\text{WM}_{\text{policy}} \wedge \neg R$. $\square$

Thus bare C, abstract-interpretation incompleteness, or acceptance incompleteness does not imply P or a weird-machine classification.

B. Defensive implications

Audit the recognized property and report separately, enumerate actor-driven operations and environment assumptions, then test report factorization. Contract violations call for implementation fixes; documented collisions call for report refinement, restriction, or isolation only when the report intended that relation.

C. Weird-machine status and a future shape theorem

The `wm_circuit` carrier supports P under its source/object and serialization premises but establishes neither R nor W. The explicitly finite M_K tuple satisfies the paper’s formal R predicate for its retrospectively declared one-step `rac_single` relation and operational prune-report;

its evidentiary label is a trace-local certificate, not a general stock-Linux report result. The auxiliary $M_{linux_r_aux_v1}$ establishes R only for its fixed custom report and restricted service semantics. Because these are different accepted artifacts and report carriers, Link forbids combining either R certificate with the interpreter's P certificate. No evaluated model therefore establishes WM_{policy} or WM_{shape} . A shape theorem must derive **macro closure** of acceptance and **report embedding** of the collision on one carrier; context sensitivity can defeat either. Complete-shell theory [22] may characterize report refinement, but cannot create reachability, reset, routing, linkage, or policy violation.

VIII. CONCLUSION

At a layered LangSec boundary, artifact recognition, post-acceptance interpretation, report factorization, bounded programmability, and policy-level weird-machine status are distinct: $R \Rightarrow C \Rightarrow A$ and $P \Rightarrow C \Rightarrow A$, while Proposition 3 gives finite countermodels for the other listed implications. The `wm_circuit` carrier records A, conditionally witnesses C, and supports P for a fixed 64-input/512-gate NAND interpreter, but establishes neither R nor W. The auxiliary tuple independently establishes R for its custom report. A separate stock-Linux capture supplies a real exact-level-0 prune event and differing common-suffix outcomes; after retrospective embedding in the declared finite `rac_single` carrier, it yields a trace-local operational-prune-relative R certificate for that tuple. It does not establish a general or Linux-specified functional-report failure. Neither R carrier supplies P or W or links to the interpreter, so the evidence establishes no WM_{shape} , policy-level weird machine, verifier unsoundness, vulnerability, or universal necessity theorem.

ETHICS AND DATA AVAILABILITY

Experiments run in an isolated local VM and use no third-party target or production data path. The interpreter runs attach to no live kernel hook. The stock-Linux R capture briefly attaches an fexit observer to verifier-internal functions during an isolated program load; it observes verifier decisions without changing the accepted program's execution. No experiment attempts corruption, verifier bypass, or privilege escalation.

The repository is <https://github.com/Emtanling/eBPF-machine>; immutable eBPF-interpreter evidence is at commit 4309069a, under `results/interpreter/interpreter-final-20260711-02/`. The auxiliary report-instance evidence is at commit f665b1a, under `results/linux_r/linux-r-v1/`. The stock-Linux trace evidence is published in the tagged V1.0 snapshot under `residuality-auditor/stock-linux-r-proof/`. Its `MANIFEST.json`, `CHECKSUMS.sha256`, embedded input hashes, raw capture, normalized certificates, proof outputs, and checker sources are co-versioned in that public snapshot. The bundled standard-library verifier emits FROZEN `_PROOF_BUNDLE_VERIFIED`. The prepublication archive `residuality-auditor-v0.3.0-full.tar.gz` had SHA-256 5fd0a2812c8c8db2fe5508440934

817c1cf9293ba0c5df31317e8b38d94a90ec; V1.0 publishes the frozen proof payload directly rather than duplicating the archive. Any frozen-byte change requires a new version, updated checksums, and re-verification.

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